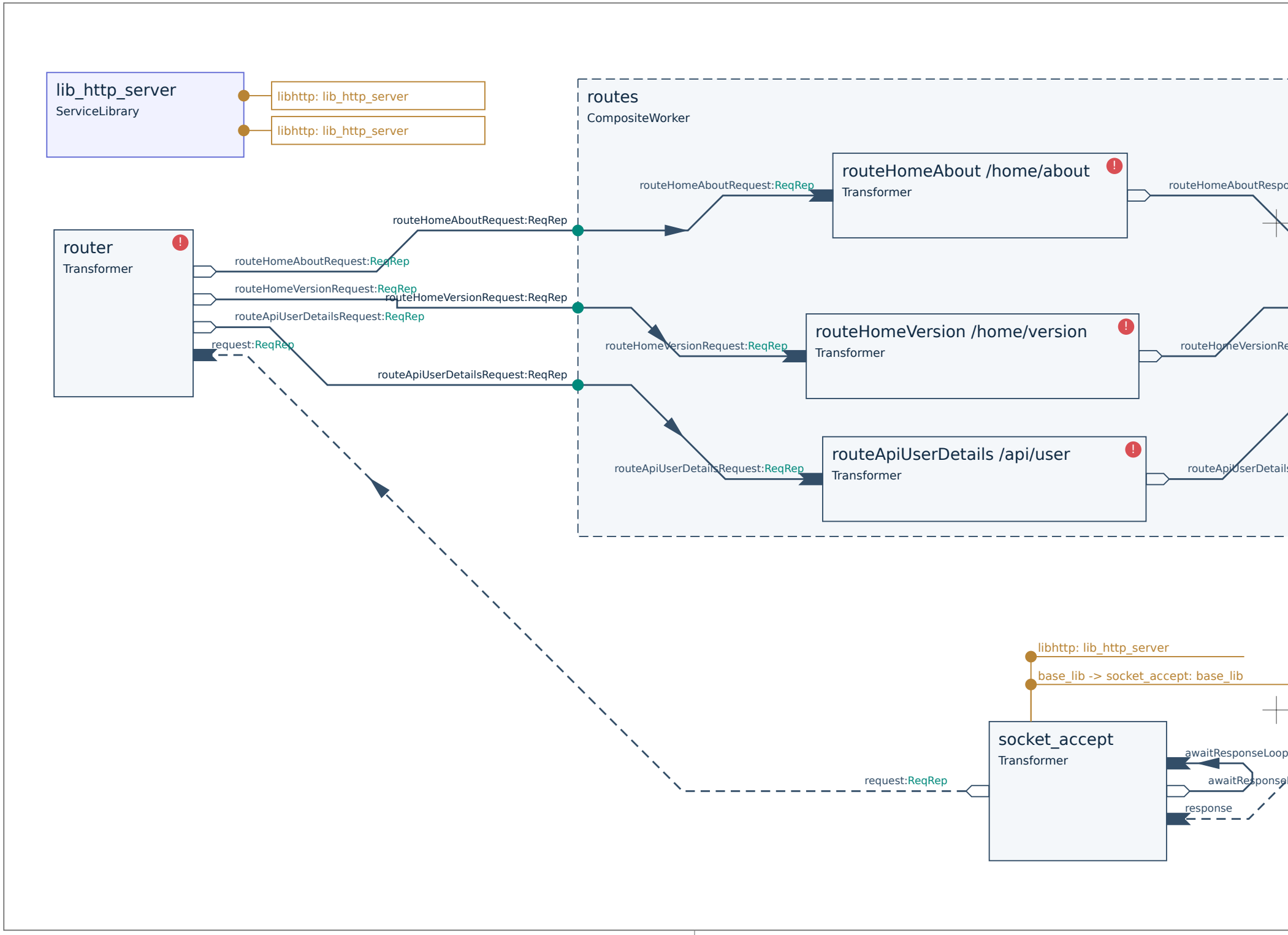
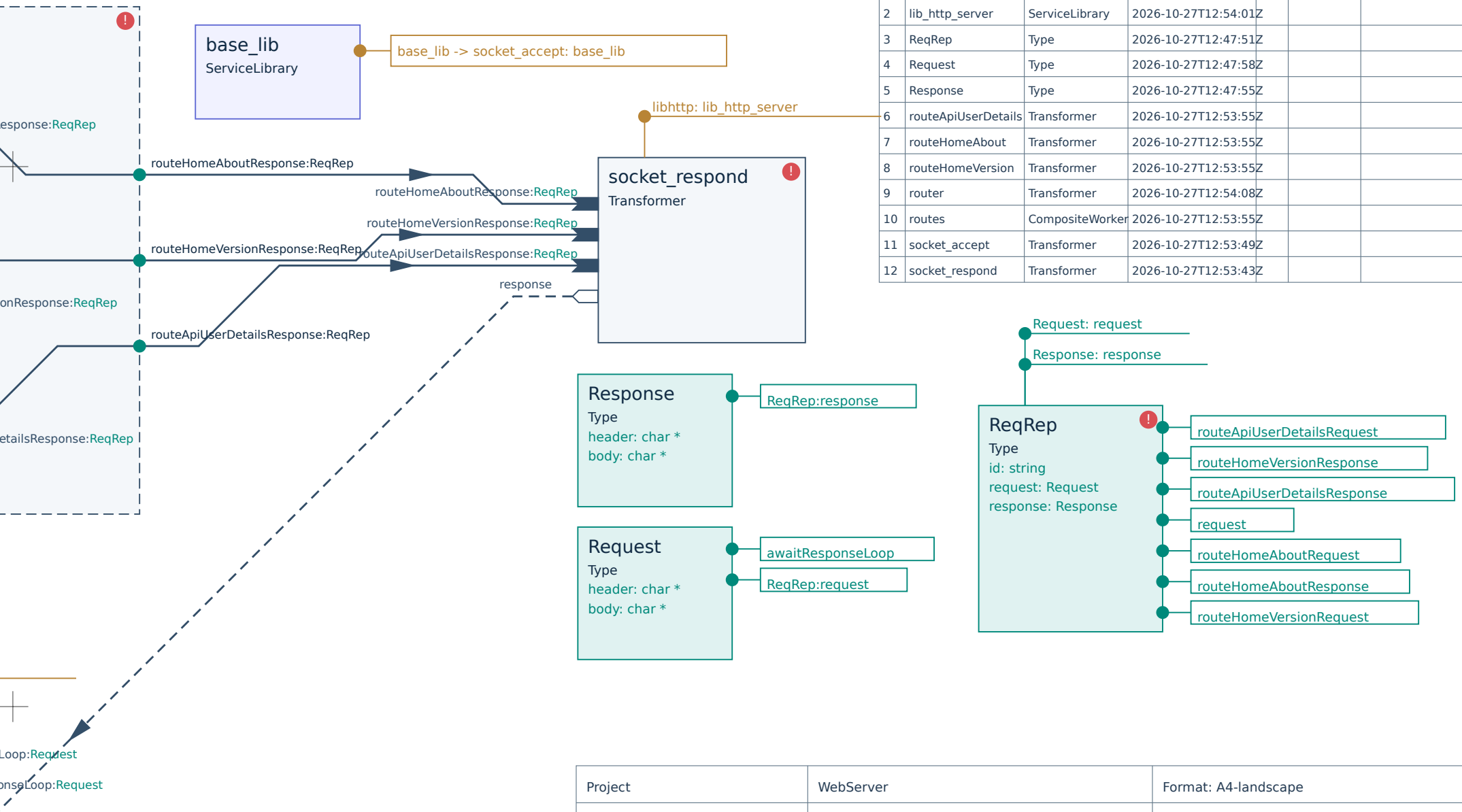

lib_http_server
ServiceLibrary
libhttp: lib_http_server
libhttp: lib_http_server
routes
CompositeWorker
routeHomeAbout /home/about
Transformer
routeHomeAboutRequest:ReqRep
routeHomeAboutRespo
router
Transformer
routeHomeAboutRequest:ReqRep
routeHomeAboutRequest:ReqRep
routeHomeVersionRequest:ReqRep
routeHomeVersionRequest:ReqRep
routeApiUserDetailsRequest:ReqRep
request:ReqRep
routeHomeVersion /home/version
Transformer
routeHomeVersionRequest:ReqRep
routeHomeVersionRe
routeApiUserDetailsRequest:ReqRep
routeApiUserDetails /api/user
Transformer
routeApiUserDetailsRequest:ReqRep
routeApiUserDetails
libhttp: lib_http_server
base_lib -> socket_accept: base_lib
socket_accept
Transformer
request:ReqRep
awaitResponseLoop
awaitResponse
response



PARTS LIST

No.	Part	Kind	Modified	Rev.	Responsible	Signature
1	base_lib	ServiceLibrary	2026-10-27T12:45:13Z			
2	lib_http_server	ServiceLibrary	2026-10-27T12:54:01Z			
3	ReqRep	Type	2026-10-27T12:47:51Z			
4	Request	Type	2026-10-27T12:47:58Z			
5	Response	Type	2026-10-27T12:47:55Z			
6	routeApiUserDetails	Transformer	2026-10-27T12:53:55Z			
7	routeHomeAbout	Transformer	2026-10-27T12:53:55Z			
8	routeHomeVersion	Transformer	2026-10-27T12:53:55Z			
9	router	Transformer	2026-10-27T12:54:08Z			
10	routes	CompositeWorker	2026-10-27T12:53:55Z			
11	socket_accept	Transformer	2026-10-27T12:53:49Z			
12	socket_respond	Transformer	2026-10-27T12:53:43Z			

Project	WebServer	Format: A4-landscape
Revision	·	Scale 1:1
Wirecraft 0.10.0 · 2 sheets		

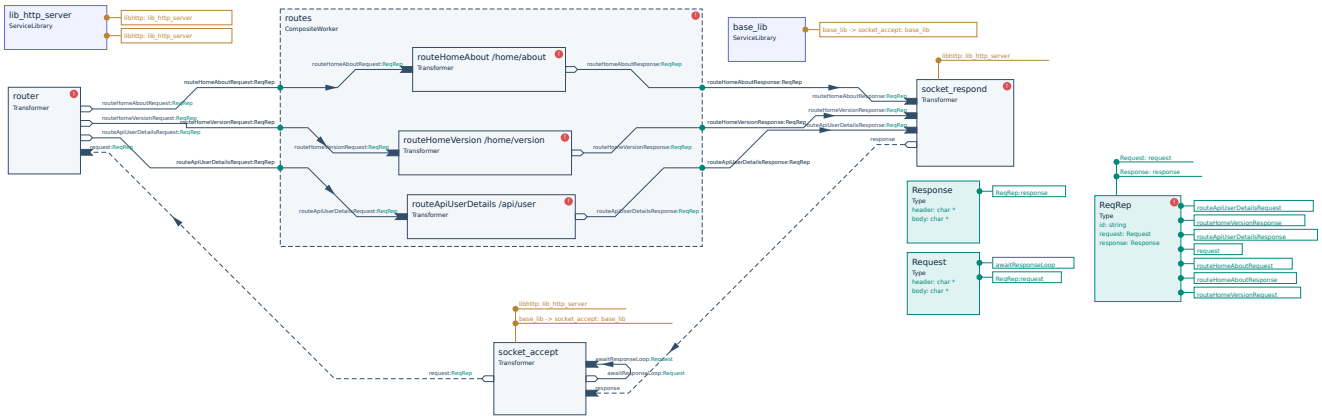
WebServer Functional and Technical Specification

Contents

- WebServer
- WebServer/socket_accept
- WebServer/router
- WebServer/Request
- WebServer/Response
- WebServer/ReqRep
- WebServer/request
- WebServer/awaitResponseLoop
- WebServer/socket_respond
- WebServer/routeHomeAboutResponse
- WebServer/routeHomeVersionResponse
- WebServer/routeApiUserDetailsResponse
- WebServer/routeHomeAboutRequest
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- WebServer/routeApiUserDetailsRequest
- WebServer/response
- WebServer/routes
- WebServer/routes/routeApiUserDetails
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- WebServer/base_lib
- WebServer/base_lib -> socket_accept
- WebServer/lib_http_server
- WebServer/libhttp
- WebServer/libhttp

Composition · WebServer

WebServer



WebServer

- ID: `root`
- Kind: Processor
- Role: CompositeWorker
- Path: WebServer
- Language: c
- Technology: c-native
- Status:
- Assignee:
- Responsible: none

socket_accept

- ID: `node_b3fdad12-aa8a-4d1f-a50c-bc43654bb903`
- Kind: Processor
- Role: Transformer
- Path: WebServer/socket_accept
- Language: c
- Technology: c-native
- Status: IMPLEMENTATION
- Assignee:
- Responsible: none

Specification

accepts incoming connections, reads the data and puts it in a new request that is pushed through the awaitResponse loop and also pushed through to the router

reads responses and matches them to requests then it responds to the awaiting client after which the request is resolved.

- Link `992f1269-5aa2-4929-91a3-5745c23f6cb9`: cf346fce-593a-4903-b9ac-51afa8814a9a:out → node_b3fdad12-aa8a-4d1f-a50c-bc43654bb903:in (auto,)
- Link `link_47dfe6fa-791b-4863-99ff-002db0ff8f88`: node_07b1aef5-8a59-4274-a70b-90995d6ea9b3:out → node_b3fdad12-aa8a-4d1f-a50c-bc43654bb903:in (data,)
- Link `link_559f3fef-7b5a-4d0e-9d55-8f70aa5c46b8`: node_edd0ae62-3c39-40da-83d6-16d04bcd1cd4:out → node_b3fdad12-aa8a-4d1f-a50c-bc43654bb903:in (lib,)
- Link `link_ff70c67f-8f63-48a5-9b9a-df23d3686dfd`: node_b3fdad12-aa8a-4d1f-a50c-bc43654bb903:out → node_b3fdad12-aa8a-4d1f-a50c-bc43654bb903:in (data,)
- Link `link_66e6e8e4-5fb1-4171-8bc9-1d22a4323ec5`: node_b3fdad12-aa8a-4d1f-a50c-bc43654bb903:out → node_afca91af-9db7-4648-894e-948214e6b2d8:in (data,)
- Link `link_ff70c67f-8f63-48a5-9b9a-df23d3686dfd`: node_b3fdad12-aa8a-4d1f-a50c-bc43654bb903:out → node_b3fdad12-aa8a-4d1f-a50c-bc43654bb903:in (data,)

router

- ID: `node_afca91af-9db7-4648-894e-948214e6b2d8`
- Kind: Processor
- Role: Transformer
- Path: WebServer/router
- Language: c
- Technology: c-native
- Status: IMPLEMENTATION
- Assignee:

- Responsible: none

Specification

receives http requests in textual form, detects their path and forwards them to the appropriate route

- Link `link_66e6e8e4-5fb1-4171-8bc9-1d22a4323ec5`: node_b3fdad12-aa8a-4d1f-a50c-bc43654bb903:out → node_afca91af-9db7-4648-894e-948214e6b2d8:in (data,)
- Link `link_1a170b1e-0115-44b9-9440-78eac48c77ce`: node_afca91af-9db7-4648-894e-948214e6b2d8:out → node_10886e3b-8eac-4654-92e5-8461794ca831:in (data,)
- Link `link_7392aa1b-0044-4e3a-a41b-84063ff15470`: node_afca91af-9db7-4648-894e-948214e6b2d8:out → node_92b69e7a-8f17-4fcb-9b2a-8e18966c9581:in (data,)
- Link `link_ea685463-77d8-4af7-87d0-b4b485ff73bc`: node_afca91af-9db7-4648-894e-948214e6b2d8:out → node_be823219-b2e9-43bf-bb78-005e6e9119e9:in (data,)

Request

- ID: `node_d65ea215-7d71-40bc-b29a-3815ca73018b`
- Kind: Type
- Role: Type
- Path: WebServer/Request
- Language: c
- Technology: c-native
- Status: TESTING
- Assignee:
- Responsible: none

Response

- ID: `node_17489b2a-30cf-4a5a-9958-bfa317dc00a2`
- Kind: Type
- Role: Type
- Path: WebServer/Response
- Language: c
- Technology: c-native
- Status: TESTING
- Assignee:
- Responsible: none

ReqRep

- ID: `node_0a2c7213-3906-4fc0-bed6-adaf8a829401`
- Kind: Type
- Role: Type
- Path: WebServer/ReqRep
- Language: c
- Technology: c-native
- Status: TESTING
- Assignee:
- Responsible: none

request

- ID: `link_66e6e8e4-5fb1-4171-8bc9-1d22a4323ec5`
- Kind: Link

- Role: Link
- Path: WebServer/request
- Language: c
- Technology: c-native
- Status:
- Assignee:
- Responsible: none

Link contract


```
{
  "sourceNodeId": "node_b3fdad12-aa8a-4d1f-a50c-bc43654bb903",
  "sourcePortName": "out",
  "targetNodeId": "node_afca91af-9db7-4648-894e-948214e6b2d8",
  "targetPortName": "in",
  "transportKind": "in-process",
  "interactionKind": "data",
  "typeName": "",
  "payloadDefinition": "",
  "typeDefinitionId": "node_0a2c7213-3906-4fc0-bed6-adaf8a829401",
  "compositeBoundaryIds": [],
  "typeExpression": null
}
```


awaitResponseLoop

- ID: `link_ff70c67f-8f63-48a5-9b9a-df23d3686dfd`
- Kind: Link
- Role: Link
- Path: WebServer/awaitResponseLoop
- Language: c
- Technology: c-native
- Status:
- Assignee:
- Responsible: none

Link contract


```
{
  "sourceNodeId": "node_b3fdad12-aa8a-4d1f-a50c-bc43654bb903",
  "sourcePortName": "out",
  "targetNodeId": "node_b3fdad12-aa8a-4d1f-a50c-bc43654bb903",
  "targetPortName": "in",
  "transportKind": "in-process",
  "interactionKind": "data",
  "typeName": "",
  "payloadDefinition": "",
  "typeDefinitionId": "node_d65ea215-7d71-40bc-b29a-3815ca73018b",
  "compositeBoundaryIds": [],
  "typeExpression": {
    "typeId": "",
    "constructorId": "array",
    "arguments": [
```

```
      {
        "typeId": "node_0a2c7213-3906-4fc0-bed6-adaf8a829401",
        "constructorId": "",
        "arguments": []
      }
    ]
  }
}
```

socket_respond

- ID: `node_07b1aef5-8a59-4274-a70b-90995d6ea9b3`
- Kind: Processor
- Role: Transformer
- Path: WebServer/socket_respond
- Language: c
- Technology: c-native
- Status: IMPLEMENTATION
- Assignee:
- Responsible: none

- Link `32e5289b-5f71-4d77-a5a5-e0a8b107bc63`: cf346fce-593a-4903-b9ac-51afa8814a9a:out → node_07b1aef5-8a59-4274-a70b-90995d6ea9b3:in (auto,)
- Link `link_1eb6b930-6980-40be-8710-8ed323817aa7`: node_be823219-b2e9-43bf-bb78-005e6e9119e9:out → node_07b1aef5-8a59-4274-a70b-90995d6ea9b3:in (data,)
- Link `link_4cc82930-d2e5-431c-a48c-8b676d08ad57`: node_10886e3b-8eac-4654-92e5-8461794ca831:out → node_07b1aef5-8a59-4274-a70b-90995d6ea9b3:in (data,)
- Link `link_d40b1933-7e41-4f24-924d-f7fd683cedfd`: node_92b69e7a-8f17-4fcb-9b2a-8e18966c9581:out → node_07b1aef5-8a59-4274-a70b-90995d6ea9b3:in (data,)
- Link `link_47dfe6fa-791b-4863-99ff-002db0ff8f88`: node_07b1aef5-8a59-4274-a70b-90995d6ea9b3:out → node_b3fdad12-aa8a-4d1f-a50c-bc43654bb903:in (data,)

routeHomeAboutResponse

- ID: `link_d40b1933-7e41-4f24-924d-f7fd683cedfd`
- Kind: Link
- Role: Link
- Path: WebServer/routeHomeAboutResponse
- Language: c
- Technology: c-native
- Status:
- Assignee:
- Responsible: none

Link contract

```
{
  "sourceNodeId": "node_92b69e7a-8f17-4fcb-9b2a-8e18966c9581",
  "sourcePortName": "out",
  "targetNodeId": "node_07b1aef5-8a59-4274-a70b-90995d6ea9b3",
  "targetPortName": "in",
  "transportKind": "in-process",
  "interactionKind": "data",
  "typeName": "",
  "payloadDefinition": "",
```

```
  "typeDefinitionId": "node_0a2c7213-3906-4fc0-bed6-adaf8a829401",
  "compositeBoundaryIds": [
    "node_f0b2f552-0e45-4843-b328-9666bb3357db"
  ],
  "typeExpression": null
}
```


routeHomeVersionResponse

- ID: `link_1eb6b930-6980-40be-8710-8ed323817aa7`
- Kind: Link
- Role: Link
- Path: WebServer/routeHomeVersionResponse
- Language: c
- Technology: c-native
- Status:
- Assignee:
- Responsible: none

Link contract


```
{
  "sourceNodeId": "node_be823219-b2e9-43bf-bb78-005e6e9119e9",
  "sourcePortName": "out",
  "targetNodeId": "node_07b1aef5-8a59-4274-a70b-90995d6ea9b3",
  "targetPortName": "in",
  "transportKind": "in-process",
  "interactionKind": "data",
  "typeName": "",
  "payloadDefinition": "",
  "typeDefinitionId": "node_0a2c7213-3906-4fc0-bed6-adaf8a829401",
  "compositeBoundaryIds": [
    "node_f0b2f552-0e45-4843-b328-9666bb3357db"
  ],
  "typeExpression": null
}
```


routeApiUserDetailsResponse

- ID: `link_4cc82930-d2e5-431c-a48c-8b676d08ad57`
- Kind: Link
- Role: Link
- Path: WebServer/routeApiUserDetailsResponse
- Language: c
- Technology: c-native
- Status:
- Assignee:
- Responsible: none

Link contract


```
{
```

```
  "sourceNodeId": "node_10886e3b-8eac-4654-92e5-8461794ca831",
  "sourcePortName": "out",
  "targetNodeId": "node_07b1aef5-8a59-4274-a70b-90995d6ea9b3",
  "targetPortName": "in",
  "transportKind": "in-process",
  "interactionKind": "data",
  "typeName": "",
  "payloadDefinition": "",
  "typeDefinitionId": "node_0a2c7213-3906-4fc0-bed6-adaf8a829401",
  "compositeBoundaryIds": [
    "node_f0b2f552-0e45-4843-b328-9666bb3357db"
  ],
  "typeExpression": null
}
```


routeHomeAboutRequest

- ID: `link_7392aa1b-0044-4e3a-a41b-84063ff15470`
- Kind: Link
- Role: Link
- Path: WebServer/routeHomeAboutRequest
- Language: c
- Technology: c-native
- Status:
- Assignee:
- Responsible: none

Link contract


```
{
  "sourceNodeId": "node_afca91af-9db7-4648-894e-948214e6b2d8",
  "sourcePortName": "out",
  "targetNodeId": "node_92b69e7a-8f17-4fcb-9b2a-8e18966c9581",
  "targetPortName": "in",
  "transportKind": "in-process",
  "interactionKind": "data",
  "typeName": "",
  "payloadDefinition": "",
  "typeDefinitionId": "node_0a2c7213-3906-4fc0-bed6-adaf8a829401",
  "compositeBoundaryIds": [
    "node_f0b2f552-0e45-4843-b328-9666bb3357db"
  ],
  "typeExpression": null
}
```


routeHomeVersionRequest

- ID: `link_ea685463-77d8-4af7-87d0-b4b485ff73bc`
- Kind: Link
- Role: Link
- Path: WebServer/routeHomeVersionRequest
- Language: c
- Technology: c-native

- Status:
- Assignee:
- Responsible: none

Link contract


```
{
  "sourceNodeId": "node_afca91af-9db7-4648-894e-948214e6b2d8",
  "sourcePortName": "out",
  "targetNodeId": "node_be823219-b2e9-43bf-bb78-005e6e9119e9",
  "targetPortName": "in",
  "transportKind": "in-process",
  "interactionKind": "data",
  "typeName": "",
  "payloadDefinition": "",
  "typeDefinitionId": "node_0a2c7213-3906-4fc0-bed6-adaf8a829401",
  "compositeBoundaryIds": [
    "node_f0b2f552-0e45-4843-b328-9666bb3357db"
  ],
  "typeExpression": null
}
```


routeApiUserDetailsRequest

- ID: `link_1a170b1e-0115-44b9-9440-78eac48c77ce`
- Kind: Link
- Role: Link
- Path: WebServer/routeApiUserDetailsRequest
- Language: c
- Technology: c-native
- Status:
- Assignee:
- Responsible: none

Link contract


```
{
  "sourceNodeId": "node_afca91af-9db7-4648-894e-948214e6b2d8",
  "sourcePortName": "out",
  "targetNodeId": "node_10886e3b-8eac-4654-92e5-8461794ca831",
  "targetPortName": "in",
  "transportKind": "in-process",
  "interactionKind": "data",
  "typeName": "",
  "payloadDefinition": "",
  "typeDefinitionId": "node_0a2c7213-3906-4fc0-bed6-adaf8a829401",
  "compositeBoundaryIds": [
    "node_f0b2f552-0e45-4843-b328-9666bb3357db"
  ],
  "typeExpression": null
}
```

response

- ID: `link_47dfe6fa-791b-4863-99ff-002db0ff8f88`
- Kind: Link
- Role: Link
- Path: WebServer/response
- Language: c
- Technology: c-native
- Status:
- Assignee:
- Responsible: none

Link contract


```
{
  "sourceNodeId": "node_07b1aef5-8a59-4274-a70b-90995d6ea9b3",
  "sourcePortName": "out",
  "targetNodeId": "node_b3fdad12-aa8a-4d1f-a50c-bc43654bb903",
  "targetPortName": "in",
  "transportKind": "in-process",
  "interactionKind": "data",
  "typeName": "",
  "payloadDefinition": "",
  "typeDefinitionId": "",
  "compositeBoundaryIds": [],
  "typeExpression": null
}
```


routes

- ID: `node_f0b2f552-0e45-4843-b328-9666bb3357db`
- Kind: Processor
- Role: CompositeWorker
- Path: WebServer/routes
- Language: c
- Technology: c-native
- Status: IMPLEMENTATION
- Assignee:
- Responsible: none

routeApiUserDetails

- ID: `node_10886e3b-8eac-4654-92e5-8461794ca831`
- Kind: Processor
- Role: Transformer
- Path: WebServer/routes/routeApiUserDetails
- Language: c
- Technology: c-native
- Status: FUNCTIONAL
- Assignee:
- Responsible: none

Specification

formulates a placehoder user data response application/json

- Link `link_1a170b1e-0115-44b9-9440-78eac48c77ce`: node_afca91af-9db7-4648-894e-948214e6b2d8:out → node_10886e3b-8eac-4654-92e5-8461794ca831:in (data,)
- Link `link_4cc82930-d2e5-431c-a48c-8b676d08ad57`: node_10886e3b-8eac-4654-92e5-8461794ca831:out → node_07b1aef5-8a59-4274-a70b-90995d6ea9b3:in (data,)

routeHomeAbout

- ID: `node_92b69e7a-8f17-4fcb-9b2a-8e18966c9581`
- Kind: Processor
- Role: Transformer
- Path: WebServer/routes/routeHomeAbout
- Language: c
- Technology: c-native
- Status: FUNCTIONAL
- Assignee:
- Responsible: none

Specification

formulates a simple about response text/plain

- Link `link_7392aa1b-0044-4e3a-a41b-84063ff15470`: node_afca91af-9db7-4648-894e-948214e6b2d8:out → node_92b69e7a-8f17-4fcb-9b2a-8e18966c9581:in (data,)
- Link `link_d40b1933-7e41-4f24-924d-f7fd683cedfd`: node_92b69e7a-8f17-4fcb-9b2a-8e18966c9581:out → node_07b1aef5-8a59-4274-a70b-90995d6ea9b3:in (data,)

routeHomeVersion

- ID: `node_be823219-b2e9-43bf-bb78-005e6e9119e9`
- Kind: Processor
- Role: Transformer
- Path: WebServer/routes/routeHomeVersion
- Language: c
- Technology: c-native
- Status: FUNCTIONAL
- Assignee:
- Responsible: none

Specification

formulates a simple version response text/plain

- Link `link_ea685463-77d8-4af7-87d0-b4b485ff73bc`: node_afca91af-9db7-4648-894e-948214e6b2d8:out → node_be823219-b2e9-43bf-bb78-005e6e9119e9:in (data,)
- Link `link_1eb6b930-6980-40be-8710-8ed323817aa7`: node_be823219-b2e9-43bf-bb78-005e6e9119e9:out → node_07b1aef5-8a59-4274-a70b-90995d6ea9b3:in (data,)

base_lib

- ID: `node_edd0ae62-3c39-40da-83d6-16d04bcd1cd4`
- Kind: Processor

- Role: ServiceLibrary
- Path: WebServer/base_lib
- Language: c
- Technology: c-native
- Status: IMPLEMENTATION
- Assignee:
- Responsible: none

Source

```
#include <stdio.h>
#include <net.h>
```

- Link `link_559f3fef-7b5a-4d0e-9d55-8f70aa5c46b8`: node_edd0ae62-3c39-40da-83d6-16d04bcd1cd4:out → node_b3fdad12-aa8a-4d1f-a50c-bc43654bb903:in (lib,)

base_lib -> socket_accept

- ID: `link_559f3fef-7b5a-4d0e-9d55-8f70aa5c46b8`
- Kind: Link
- Role: Link
- Path: WebServer/base_lib -> socket_accept
- Language: c
- Technology: c-native
- Status:
- Assignee:
- Responsible: none

Link contract


```
{
  "sourceNodeId": "node_edd0ae62-3c39-40da-83d6-16d04bcd1cd4",
  "sourcePortName": "out",
  "targetNodeId": "node_b3fdad12-aa8a-4d1f-a50c-bc43654bb903",
  "targetPortName": "in",
  "transportKind": "in-process",
  "interactionKind": "lib",
  "typeName": "",
  "payloadDefinition": "",
  "typeDefinitionId": "",
  "compositeBoundaryIds": [],
  "typeExpression": null
}
```


lib_http_server

- ID: `cf346fce-593a-4903-b9ac-51afa8814a9a`
- Kind: Processor
- Role: ServiceLibrary
- Path: WebServer/lib_http_server
- Language: c
- Technology: c-native
- Status: BUSINESS

- Assignee:
- Responsible: none

Specification

contains all the necessary functions to support accepting and reading requests, initializing a ReqRep object, getting the response and using the response to handle the response to the client.

- Link `32e5289b-5f71-4d77-a5a5-e0a8b107bc63`: cf346fce-593a-4903-b9ac-51afa8814a9a:out → node_07b1aef5-8a59-4274-a70b-90995d6ea9b3:in (auto,)
- Link `992f1269-5aa2-4929-91a3-5745c23f6cb9`: cf346fce-593a-4903-b9ac-51afa8814a9a:out → node_b3fdad12-aa8a-4d1f-a50c-bc43654bb903:in (auto,)

libhttp

- ID: `992f1269-5aa2-4929-91a3-5745c23f6cb9`
- Kind: Link
- Role: Link
- Path: WebServer/libhttp
- Language: c
- Technology: c-native
- Status:
- Assignee:
- Responsible: none

Link contract


```
{
  "sourceNodeId": "cf346fce-593a-4903-b9ac-51afa8814a9a",
  "sourcePortName": "out",
  "targetNodeId": "node_b3fdad12-aa8a-4d1f-a50c-bc43654bb903",
  "targetPortName": "in",
  "transportKind": "in-process",
  "interactionKind": "auto",
  "typeName": "",
  "payloadDefinition": "",
  "typeDefinitionId": "",
  "compositeBoundaryIds": [],
  "typeExpression": null
}
```


libhttp

- ID: `32e5289b-5f71-4d77-a5a5-e0a8b107bc63`
- Kind: Link
- Role: Link
- Path: WebServer/libhttp
- Language: c
- Technology: c-native
- Status:
- Assignee:
- Responsible: none

Link contract


```
{
  "sourceNodeId": "cf346fce-593a-4903-b9ac-51afa8814a9a",
  "sourcePortName": "out",
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  "targetPortName": "in",
  "transportKind": "in-process",
  "interactionKind": "auto",
  "typeName": "",
  "payloadDefinition": "",
  "typeDefinitionId": "",
  "compositeBoundaryIds": [],
  "typeExpression": null
}
```